

Chapter 6

Conditioning and Learning



Types of Learning

Learning: relatively permanent change in behavior due to experience

- Does NOT include temporary changes due to disease, fatigue, injury, maturation, or drugs

Associative learning: the formation of simple associations among stimuli and/or responses – two types Classical and Operant Conditioning

Cognitive learning: understanding, knowing, anticipating, or otherwise making use of information-rich higher mental processes



Associative Learning

- **Reinforcement:** any event that increases the probability that a response will recur
- **Antecedents:** events that precede a response; more important in classical conditioning
- **Consequences:** effects that follow a response; more important in operant conditioning

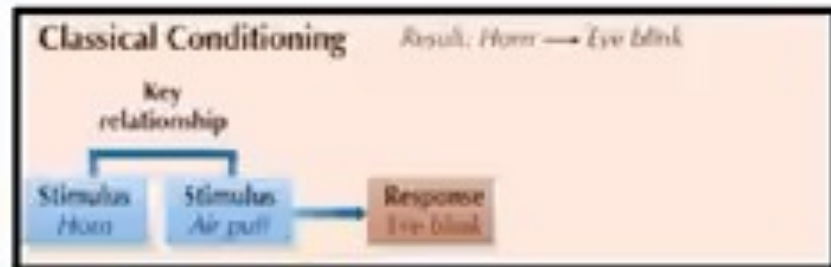
Classical
Conditioning



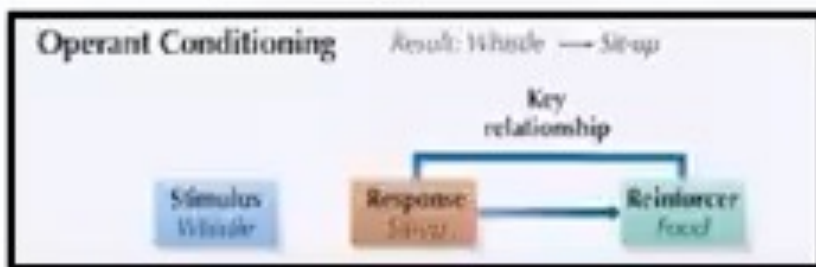
Operant
Conditioning

Classical vs. Operant Conditioning

Classical conditioning: the process by which an antecedent stimulus that doesn't produce a response is linked with one that does (a horn is associated with a puff of air)



Operant conditioning: the process by which the likelihood of a response is influenced by the consequences of responding



Classical Conditioning

Discovered by Ivan Pavlov, the Russian physiologist who studied digestion by presenting dogs with meat powder and measuring salivation



YouTube Video

1 - Classical Conditioning
Pavlov explanation
<http://www.youtube.com/watch?v=hhqumfpxuzl>

Pavlovian Terms

Neutral stimulus: stimulus that does not evoke a response

Conditioned stimulus (CS): stimulus that evokes a response because it has been repeatedly paired with an unconditioned stimulus

Unconditioned stimulus (US): stimulus innately capable of eliciting a response

Unconditioned response (UR): innate reflex response elicited by an unconditioned stimulus (US)

Conditioned response (CR): learned response elicited by a conditioned stimulus

Definition of *PAVLOVIAN*

- 1 : of or relating to Ivan *Pavlov* or to his work and theories <*Pavlovian* conditioning>
- 2 : being or expressing a conditioned or predictable reaction : *Pavlovian* <the candidates gave *Pavlovian* answers>

Extinction & Spontaneous Recovery

YouTube Video

2 - Classical Conditioning - Pavlov Demonstrated (Funny)

<https://www.youtube.com/watch?v=U3C0Dm8l5sU>

Extinction: weakening of a conditioned response through removal of reinforcement

Spontaneous recovery: reappearance of a learned response following apparent extinction



Classical Conditioning in Humans

Phobia: fear that persists even when no realistic danger exists (e.g., arachnophobia; fear of spiders)

YouTube Video
3 - Classical
Conditioning
Little Albert
Experiment

<http://www.youtube.com/watch?v=9h8frXACsOI&feature=related>



Desensitization: decreasing fear or anxiety by exposing phobic people gradually to feared stimuli while they stay calm and relaxed

Operant Conditioning

Learning is based on the consequences of responding;
we associate responses with their consequences

- **Law of effect**: the probability of a response is altered by the effect it has, so responses that lead to desired effects are repeated; those that lead to undesired effects are not
- **Reinforcer - Reinforcement**: any event that follows a response and increases its likelihood of recurring – an increase in behavior
- **Punisher - Punishment**: any consequence that reduces the frequency of a target behavior – a decrease in behavior

YouTube Video

4 - Operant Conditioning

Skinner's Pigeons

a little slow but good detail

<http://www.youtube.com/watch?v=yhvaSEJtOV8&feature=related>

Two Types of Reinforcement

- **Positive reinforcement:** when a response is followed by a reward or other positive event
- **Negative reinforcement:** when a response is followed by the removal of an unpleasant event (e.g., the bells in Fannie's car stop when she puts the seatbelt on); ends discomfort



YouTube Video

5 - Positive and Negative Reinforcement (funny)

<http://www.youtube.com/watch?v=K2uPCMPA8I0&feature=related>

Timing of Reinforcement

Operant reinforcement most effective when given immediately after a correct response

Shaping: molding responses gradually in a step-by-step fashion to a desired pattern

Superstitious behaviors: behaviors that are repeated because they appear to produce reinforcement (i.e., the timing is right), even though they are not necessary



[YouTube Video](#)

6 - Operant Conditioning - Review

Punishment

Punishment: any *aversive consequence* that follows a response and decreases the likelihood of it recurring (e.g., a spanking)

Keys: timing, consistency, and intensity

- **Severe punishment:** intense punishment, capable of suppressing a response for a long period
- **Mild punishment:** weak punishment; usually slows responses temporarily



Downside of Punishment

Aversive stimulus: stimulus that is painful or uncomfortable (e.g., a shock)

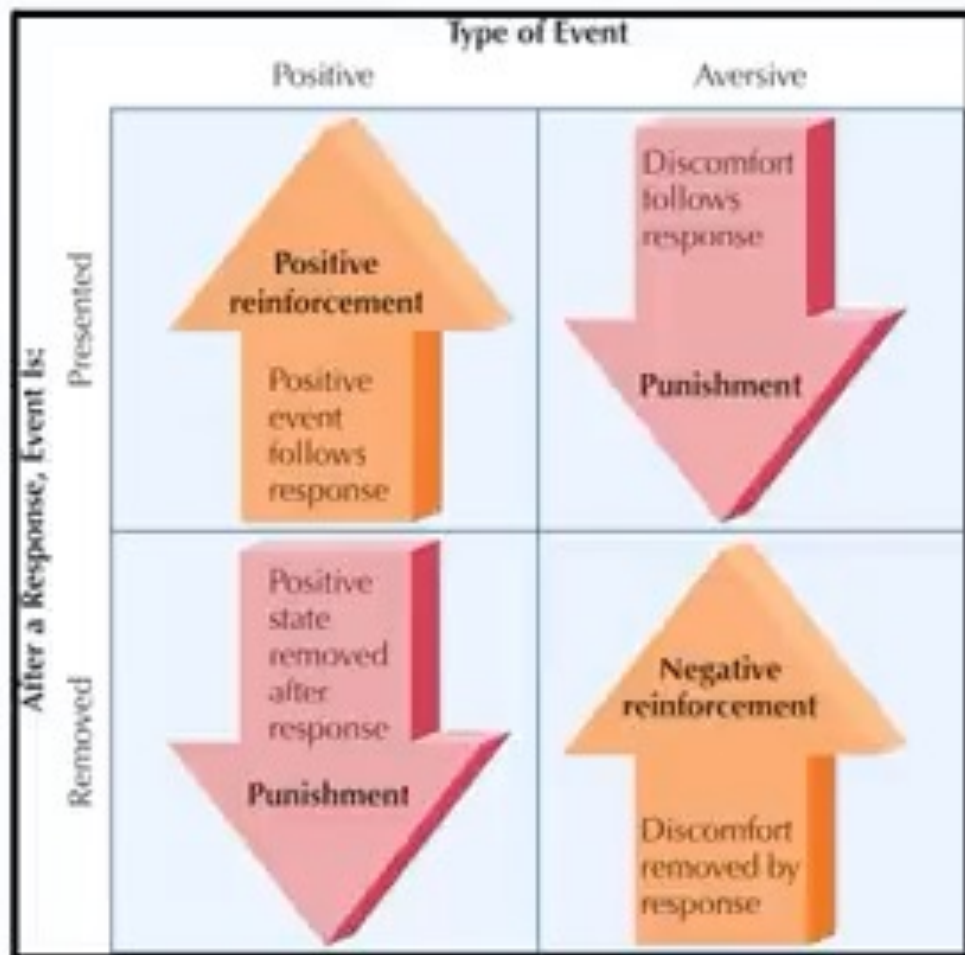
Escape learning: learning to make a response to end an aversive stimulus

Avoidance learning: learning to make a response to avoid, postpone, or prevent discomfort (e.g., not going to a doctor or dentist)

Punishment may also increase aggression



Operant Conditioning



Operant Reinforcers

Primary reinforcer: unlearned and natural; satisfies physiological needs (food, water, sex)



Secondary reinforcer: learned reinforcer (money, grades, approval, praise); gains reinforcing properties by associating with a primary reinforcer

Token reinforcer: tangible secondary reinforcer



The End

